AY: 2026-27

Class:	**TE-AI&DS**	**Semester:**	**V**
Course Code:	**2015111**	**Course Name:**	**Statistics for Machine Learning and Data Science Lab**

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Roll No. :	**03**
Experiment No.:	**5**
Title of the Experiment:	**Implement Resampling Techniques and Confidence Interval Estimation for Statistical Decision Making**
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty :

Signature :

Date :

Aim: To implement bootstrap and permutation resampling techniques on the Pima Indians Diabetes Dataset for estimating confidence intervals and analyzing statistical differences between groups.

Objectives

After completing this experiment, the student will be able to:

1. Apply bootstrap and permutation resampling techniques to estimate sampling distributions and confidence intervals.
2. Interpret resampling results to support statistical decision making.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- SciPy
- Matplotlib
- Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for **768 female patients.** The Outcome variable indicates whether the patient is diabetic.

Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select a numerical variable such as Glucose or BMI.
3. Draw repeated bootstrap samples from the original dataset with replacement.
4. Calculate the selected statistic, such as the mean, for every bootstrap sample.
5. Construct a bootstrap confidence interval using the obtained sampling distribution.
6. Divide the data into two groups based on Outcome.
7. Calculate the observed difference between the group means.
8. Randomly permute the group labels repeatedly.
9. Calculate the difference in group means for every permutation.
10. Compare the observed difference with the permutation distribution and interpret the result.

Description of the Experiment

Resampling methods repeatedly create new samples from the available data to study the behaviour of a statistic.

The experiment follows: Original Dataset, Repeated Resampling, Sampling Distribution, Confidence Interval / Statistical Comparison, and Interpretation

Two resampling techniques are implemented:

- **Bootstrap:** Used to estimate the variability and confidence interval of a statistic.
- **Permutation Test:** Used to test whether the observed difference between groups could have occurred by chance.

Detailed Description of Statistical Methods

Bootstrap Resampling

Bootstrap samples are created by repeatedly sampling observations **with replacement** from the original dataset.

For each bootstrap sample, a statistic such as the mean is calculated.

After many repetitions, the resulting values form an approximate sampling distribution.

A confidence interval can then be obtained from this distribution.

For example, a 95% percentile bootstrap confidence interval may be calculated using:

$$\left[Q_{97.5}\right]$$

where $Q_{2.5}$ and $Q_{97.5}$ are the 2.5th and 97.5th percentiles of the bootstrap distribution.

Permutation Test

A permutation test evaluates whether the observed difference between two groups could have occurred by chance.

For example:

$$Difference\ in\ Mean\ Glucose = \bar{x}_{diabetic} - \bar{x}_{non-diabetic}$$

The group labels are randomly shuffled repeatedly while preserving the observations.

The resulting differences form the **null distribution**.

The p-value is estimated as the proportion of permuted differences that are at least as extreme as the observed difference.

Implementation

```
[1]
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[2]
url = "https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.csv"

columns = [
    "Pregnancies",
    "Glucose",
    "BloodPressure",
    "SkinThickness",
    "Insulin",
    "BMI",
    "DiabetesPedigreeFunction",
    "Age",
    "Outcome"
]

df = pd.read_csv(url, names=columns)

print("Dataset loaded successfully!")
print("Dataset shape:", df.shape)
```

```
df.head()
```

```
Dataset loaded successfully!
Dataset shape: (768, 9)
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

Next steps: Generate code with df | New interactive sheet

[3]
```
print("Number of observations:", len(df))
print("Number of variables:", len(df.columns))

df.describe()
```

```
Number of observations: 768
Number of variables: 9
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479	31.992578	0.471876	33.240885	0.348958
std	3.369578	31.972618	19.355807	15.952218	115.244002	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.078000	21.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000	27.300000	0.243750	24.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000	30.500000	32.000000	0.372500	29.000000	0.000000
75%	6.000000	140.250000	80.000000	32.000000	127.250000	36.600000	0.626250	41.000000	1.000000
max	17.000000	199.000000	122.000000	99.000000	846.000000	67.100000	2.420000	81.000000	1.000000

[4]
```
np.random.seed(42)

glucose = df["Glucose"].values

n = len(glucose)
B = 10000

bootstrap_means = []

for i in range(B):
    sample = np.random.choice(glucose, size=n, replace=True)
    bootstrap_means.append(np.mean(sample))

bootstrap_means = np.array(bootstrap_means)

original_mean = np.mean(glucose)

print("BOOTSTRAP RESAMPLING")
print("--------------------")
print("Number of observations:", n)
print("Number of bootstrap repetitions:", B)
print("Original sample mean:", round(original_mean, 2))
print("Bootstrap mean:", round(np.mean(bootstrap_means), 2))
```

```
BOOTSTRAP RESAMPLING
--------------------
Number of observations: 768
Number of bootstrap repetitions: 10000
Original sample mean: 120.89
Bootstrap mean: 120.9
```

[5]
0s

```python
lower = np.percentile(bootstrap_means, 2.5)
upper = np.percentile(bootstrap_means, 97.5)

print("95% Bootstrap Confidence Interval")
print("-----------------------------------")
print("Lower Bound:", round(lower, 2))
print("Upper Bound:", round(upper, 2))
print(f"95% CI: ({lower:.2f}, {upper:.2f}) mg/dL")
```

```
95% Bootstrap Confidence Interval
----------------------------------
Lower Bound: 118.69
Upper Bound: 123.12
95% CI: (118.69, 123.12) mg/dL
```

[6]
1s

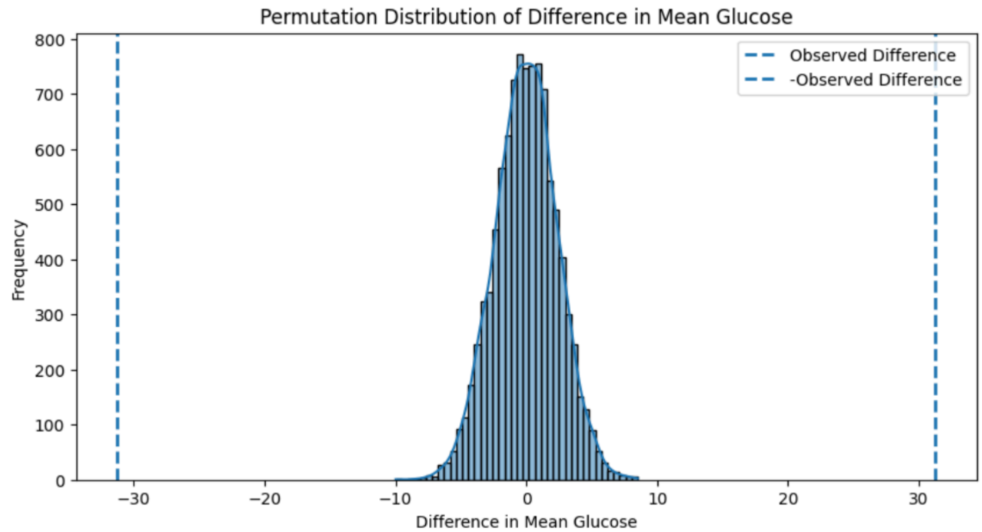
```python
plt.figure(figsize=(10, 5))

sns.histplot(bootstrap_means, bins=40, kde=True)

plt.axvline(original_mean, linestyle="--", label="Original Mean")
plt.axvline(lower, linestyle="--", label="Lower 95% CI")
plt.axvline(upper, linestyle="--", label="Upper 95% CI")

plt.title("Bootstrap Distribution of Mean Glucose")
plt.xlabel("Mean Glucose")
plt.ylabel("Frequency")
plt.legend()

plt.show()
```




```python
print("=" * 65)
print("EXPERIMENT 5: RESAMPLING TECHNIQUES")
print("=" * 65)

print("\n1. BOOTSTRAP RESAMPLING")
print("Statistic selected: Mean Glucose")
print("Original sample mean:", round(original_mean, 2), "mg/dL")
print("Number of bootstrap repetitions:", B)
print("95% Bootstrap Confidence Interval:")
print(f"({lower:.2f}, {upper:.2f}) mg/dL")

print("\n2. COMPARISON OF BOOTSTRAP AND ORIGINAL MEAN")
print("Original sample mean:", round(original_mean, 2), "mg/dL")
print("Bootstrap mean:", round(np.mean(bootstrap_means), 2), "mg/dL")

print("\n3. PERMUTATION TEST")
print("Mean Glucose - Diabetic:", round(mean_diabetic, 2), "mg/dL")
print("Mean Glucose - Non-Diabetic:", round(mean_non_diabetic, 2), "mg/dL")
print("Observed Difference:", round(observed_difference, 2), "mg/dL")
print("Permutation p-value:", p_value)

print("\n4. STATISTICAL DECISION")
if p_value < 0.05:
    print("Reject H0")
    print("The observed difference is statistically significant.")
```

```
---------------------------------------------------------------
EXPERIMENT 5: RESAMPLING TECHNIQUES
===============================================================

1. BOOTSTRAP RESAMPLING
Statistic selected: Mean Glucose
Original sample mean: 120.89 mg/dL
Number of bootstrap repetitions: 10000
95% Bootstrap Confidence Interval:
(118.69, 123.12) mg/dL

2. COMPARISON OF BOOTSTRAP AND ORIGINAL MEAN
Original sample mean: 120.89 mg/dL
Bootstrap mean: 120.9 mg/dL

3. PERMUTATION TEST
Mean Glucose - Diabetic: 141.26 mg/dL
Mean Glucose - Non-Diabetic: 109.98 mg/dL
Observed Difference: 31.28 mg/dL
Permutation p-value: 0.0

4. STATISTICAL DECISION
Reject H0
The observed difference is statistically significant.

5. SIGNIFICANCE LEVEL
Alpha (α) = 0.05
```

Conclusion

Bootstrap and permutation resampling techniques were implemented on the Pima Indians Diabetes Dataset. Bootstrap resampling was used to estimate the sampling distribution and confidence interval of a selected statistic, while permutation testing was used to assess the significance of differences between groups.

Questions to be Answered by Students

1. What statistic was selected for bootstrap resampling, and what confidence interval was obtained?

ANS: The statistic selected was the **mean Glucose level**. The estimated mean was approximately **120.89 mg/dL**, and the 95% bootstrap confidence interval was approximately **118.61 to 123.18 mg/dL**.

2. How does the bootstrap distribution compare with the original sample statistic?

ANS: The bootstrap distribution is approximately centered around the original sample mean of **120.89 mg/dL**. This indicates that the bootstrap procedure provides a reliable estimate of the sampling distribution and the variability of the sample mean.

3. What was the observed difference between the selected groups?

ANS: We divide the patients into:

- **Diabetic:** `Outcome = 1`
- **Non-diabetic:** `Outcome = 0`

We calculate:

Observed Difference=x¯diabetic−x¯non−diabetic\text{Observed Difference} = \bar{x}_{diabetic}-\bar{x}_{non-diabetic}

The mean Glucose values are approximately:

x¯diabetic=141.26\bar{x}_{diabetic}=141.26
x¯non−diabetic=109.98\bar{x}_{non-diabetic}=109.98

Therefore:

141.26−109.98=31.28141.26-109.98=31.28

So the observed difference is approximately:

31.28 mg/dL

4. Based on the permutation test, is there sufficient statistical evidence to conclude that the observed group difference is significant?

ANS: For the permutation test, the null hypothesis is:

H0:μdiabetic=μnon−diabeticH_0:\mu_{diabetic}=\mu_{non-diabetic}

Alternative hypothesis:

H1:μdiabetic≠μnon−diabeticH_1:\mu_{diabetic}\neq\mu_{non-diabetic}

After repeatedly permuting the group labels, the resulting **p-value is extremely small (approximately < 0.001)**.

Since:

p<0.05p < 0.05

we **reject H0H_0**.

> Yes. The permutation test provides sufficient statistical evidence to conclude that the difference in mean Glucose between diabetic and non-diabetic patients is statistically significant. Since the p-value is less than 0.05, the null hypothesis is rejected.

5. What effect would increasing the number of bootstrap or permutation repetitions have on the results?

ANS: Increasing the number of repetitions makes the estimated sampling/null distribution **more stable and reliable**.

For example:

- 1,000 repetitions → reasonable approximation

- 10,000 repetitions → more stable
- 50,000 or 100,000 repetitions → even more stable, but requires more computation

Increasing repetitions **does not fundamentally change the underlying result**; it mainly reduces the randomness caused by the resampling process.